Name:_____

Exam 3 - MATH-262-02

Instructions: You have 110 minutes to take this exam. Please answer each question in the space provided. Answers without justification may ***earn no credit***. You are permitted your signed prep work journal and a calculator on this exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

1. Use the ***definition*** of the Laplace transform to compute $\mathcal{L}\{f\}$ where

$$f(t) = \begin{cases} t & 0 \leq t < 2; \\ 4 & t \geq 4. \end{cases}$$

2. Compute

$$\mathcal{L}^{-1}\left\{\frac{9+s}{4-s^2}\right\}.$$

3. Use Laplace transforms to solve the IVP

$$u'' + u = \cos(3t);\ u(0) = 1,\ u'(0) = 0.$$

4. Find two power series solutions to the ODE

$$(x - 3)y' + 2y = 0$$

that are centered at $x = 0$

5. Consider the ODE

$$2x^2(1+x)y'' + 3x(1+x)^3y' - (1-x^2)y = 0.$$

(a) Show that $x = 0$ is a regular singular point for the ODE.

(b) Show that the equation must have Frobenius indeces of $r = \frac{1}{2}, -1$.